The gradient of L is then given by

$$\begin{aligned}
\nabla_W L(W) &= \mathbb{E}_{(z,y)}\left[\sum_{k=1}^{N} \frac{\partial \ell}{\partial \xi_k}(y, W_U W w_E(z)) \nabla_W (w_U(k)^\top W w_E(z))\right] \\
&= \mathbb{E}_{(z,y)}\left[\sum_{k=1}^{N} (\hat{p}_W(k|z) - \mathbb{1}\{y=k\}) w_U(k) w_E(z)^\top\right] \\
&= \sum_{k=1}^{N} \mathbb{E}_z[\mathbb{E}_y[(\hat{p}_W(k|z) - \mathbb{1}\{y=k\}) w_U(k) w_E(z)^\top \mid z]] \\
&= \sum_{k=1}^{N} \mathbb{E}_z[(\hat{p}_W(k|z) - \mathbb{E}_y[\mathbb{1}\{y=k\}|z]) w_U(k) w_E(z)^\top],
\end{aligned}$$

which yields the desired result. □

C.2 Proof of Lemma 2

Proof. Using similar steps as the proof of Lemma 1, we have

$$\begin{aligned}
\nabla_W L(W) &= \mathbb{E}_{(x,y)}\left[\sum_{k=1}^{N} \frac{\partial \ell}{\partial \xi_k}(y, W_U W x) \nabla_W (w_U(k)^\top W x)\right] \\
&= \mathbb{E}_{(x,y)}\left[\sum_{k=1}^{N} (\hat{p}_W(k|x) - \mathbb{1}\{y=k\}) w_U(k) x^\top\right] \\
&= \sum_{k=1}^{N} w_U(k)\, \mathbb{E}_x[\hat{p}_W(k|x) x]^\top - \sum_{k=1}^{N} \mathbb{E}_y[\mathbb{1}\{y=k\} w_U(k)\, \mathbb{E}[x|y]^\top] \\
&= \sum_{k=1}^{N} w_U(k)\, \mathbb{E}_x[\hat{p}_W(k|x) x]^\top - \sum_{k,j=1}^{N} p(y=j)\, \mathbb{1}\{j=k\} w_U(k)\, \mathbb{E}[x|y=j]^\top \\
&= \sum_{k=1}^{N} p(y=k) w_U(k) (\hat{\mu}_k - \mu_k)^\top,
\end{aligned}$$

with $\hat{\mu}_k = p(y=k)^{-1}\, \mathbb{E}_x[\hat{p}_W(k|x) x]$ and $\mu_k = \mathbb{E}[x|y=k]$. □

C.3 Proof of Lemma 4

Proof. To better isolate the role of keys from values, we denote the keys that are fed into the matrix W by $Z = [z_1, \ldots, z_T] \in \mathbb{R}^{d \times T}$, while the query is simply x_T. In practice we have $Z = X$, and both are superpositions of potentially multiple embeddings (if W is part of the second attention layer, these are the token embedding, positional embedding, and the output of the first attention layer).

The gradient of the loss at $W = 0$ writes:

$$\nabla_W L(W)\big|_{W=0} = \mathbb{E}_{(X,Z,y)}\left[\sum_{k=1}^{N} \frac{\partial \ell}{\partial \xi_k}(y, \xi) \cdot \nabla_W (w_U(k)^\top \Phi_2 X \sigma(Z^\top W x_T))\big|_{W=0}\right] \tag{30}$$

$$= \mathbb{E}_{(X,Z,y)}\left[\sum_{k=1}^{N} (\hat{p}_W(k|X,Z) - \mathbb{1}\{y=k\}) \cdot \nabla_W (w_U(k)^\top \Phi_2 X \sigma(Z^\top W x_T))\big|_{W=0}\right]. \tag{31}$$

We have

$$\begin{aligned}
\nabla_W (w_U(k)^\top \Phi_2 X \sigma(Z^\top W x_T))\big|_{W=0} &= \sum_{t=1}^{T} w_U(k)^\top \Phi_2 x_t \cdot \nabla_W (\sigma(Z^\top W x_T)_t) \\
&= \frac{1}{T} \sum_{t=1}^{T} w_U(k)^\top \Phi_2 x_t \cdot (z_t - \bar{z}_{1:T}) x_T^\top,
\end{aligned}$$

where $\bar{z}_{1:T} = \frac{1}{T}\sum_t z_t$, and we used the fact that

$$\frac{\partial}{\partial u_s}\sigma(u)_t\big|_{u=0} = \frac{1}{T}\,\mathbb{1}\{t=s\} - \frac{1}{T^2}. \tag{32}$$

The gradient (31) now writes

$$\nabla_W L(W)\big|_{W=0} = \sum_{k=1}^{N}\mathbb{E}_{(X,Z)}[(\hat{p}_W(k|X,Z) - \mathbb{1}\{y=k\})\frac{1}{T}\sum_{t=1}^{T} w_U(k)^\top \Phi_2 x_t \cdot (z_t - \bar{z}_{1:T})x_T^\top],$$

and the result follows. □

C.4 Proof of Lemma 5

Proof. The linearization of the second layer softmax around zero takes the following form:

$$\bar{\sigma}(Z^\top W_2 x_T)_t = \frac{1}{T}(1 + z_t^\top W_2 x_T - \bar{z}_{1:T}^\top W_2 x_T),$$

with $z_t = \sum_{s=1}^{t}\Phi_1 x_s \sigma(p_{1:t}^\top W p_t)_s$ the output of the first attention layer.

$$\begin{aligned}
\xi_k &= \sum_{t=1}^{T} w_U(k)^\top \Phi_2 x_t \bar{\sigma}(Z^\top W_2 x_T) \\
&= \frac{1}{T}\sum_{t=1}^{T} w_U(k)^\top \Phi_2 x_t + \frac{1}{T}\sum_{t=1}^{T} w_U(k)^\top \Phi_2 x_t \sum_{s=1}^{t}(\Phi_1 x_s)^\top W_2 x_T \sigma(p_{1:t}^\top W p_t)_s \\
&\quad - w_u(k)^\top \Phi_2 \bar{x}_{1:T} \cdot \frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{t}(\Phi_1 x_s)^\top W_2 x_T \sigma(p_{1:t}^\top W p_t)_s.
\end{aligned}$$

Then,

$$\nabla_W L(W)\big|_{W=0} \tag{33}$$

$$= \mathbb{E}_{(X,y)}\left[\sum_{k=1}^{N}\frac{\partial \ell}{\partial \xi_k}(y,\xi)\cdot \nabla_W \xi_k\big|_{W=0}\right] \tag{34}$$

$$= \mathbb{E}_{(X,y)}\left[\sum_{k=1}^{N}\frac{\partial \ell}{\partial \xi_k}(y,\xi)\frac{1}{T}\sum_{t=1}^{T} w_U(k)^\top \Phi_2 x_t \frac{1}{t}\sum_{s=1}^{t}(\Phi_1 x_s)^\top W_2 x_T (p_s - \bar{p}_{1:t})p_t^\top\right] \tag{35}$$

$$- \mathbb{E}_{(X,y)}\left[\sum_{k=1}^{N}\frac{\partial \ell}{\partial \xi_k}(y,\xi) w_U(k)^\top \Phi_2 \bar{x}_{1:T} \cdot \frac{1}{T}\sum_{t=1}^{T}\frac{1}{t}\sum_{s=1}^{t}(\Phi_1 x_s)^\top W_2 x_T (p_s - \bar{p}_{1:t})p_t^\top\right], \tag{36}$$

using (32). The result follows by using $\frac{\partial \ell}{\partial \xi_k}(y,\xi) = \hat{p}(k|\xi) - \mathbb{1}\{y=k\}$.

□

D Beyond our Simplified Architecture

While the simplified architecture presented in Section 4.2 is sufficient to capture the desired induction behavior for our bigram task, transformer architectures used in practice typically involve more components, as well as more heads and layers. In this section, we discuss how our memory viewpoint extends to such architectures.

Factorizations. In practice, transformers typically involve products of matrices, potentially with a low-rank bottleneck. For instance, our key-query matrices W_K should instead be considered as a product $W_K^\top W_Q$, and the output-value matrices W_O and W_V are typically jointly optimized.

Consider an associative memory of the form:

$$W_* = \sum_i y_i x_i^\top \in \mathbb{R}^{d\times d},$$

where $(x_i)_i$ and $(y_i)_i$ are appropriate collections of near-orthonormal embeddings.

We now argue that a similar associative memory can be achieved with the factorization $W = \frac{d}{2d'}UV$, where $U \in \mathbb{R}^{d\times d'}$, $V \in \mathbb{R}^{d'\times d}$ with $d' \le d$ (for instance d' could be the dimension of attention heads), are given by:[3]

$$U = U_0 + \sum_i y_i (V_0 x_i)^\top$$
$$V = V_0 + \sum_i (U_0^\top y_i) x_i^\top,$$

where U_0 and V_0 are random matrices with $\mathcal{N}(0, \frac{1}{d})$ entries. These matrices are similar to those that would arise from a single gradient step individually on U and V from initializations U_0 and V_0, as in Lemma 1. To see why W behaves like W_*, note that we have

$$UV = U_0V_0 + \sum_i y_i (V_0x_i)^\top V_0 + \sum_i (U_0^\top y_i)x_i^\top + \sum_{i,j} y_i (V_0x_i)^\top (U_0^\top y_j) x_j^\top.$$

It is also easy to check using central limit arguments (similar to remapping in Appendix A) that $\tilde{x}_i := \sqrt{\frac{d}{d'}} V_0 x_i \in \mathbb{R}^{d'}$ and $\tilde{y}_i := \sqrt{\frac{d}{d'}} U_0^\top y_i$ are all nearly-orthonormal embeddings. Thus, we have

$$\begin{aligned} 2y_k^\top W x_l &= \tilde{y}_k^\top \tilde{x}_l + \sum_i y_k^\top y_i \tilde{x}_i^\top \tilde{x}_l + \sum_i \tilde{y}_k^\top \tilde{y}_i x_i^\top x_l + \sum_{i,j} y_k^\top y_i \tilde{x}_i^\top \tilde{y}_j x_j^\top x_l \\ &\approx 0 + \mathbb{1}\{k=l\} + \mathbb{1}\{k=l\} + 0, \end{aligned}$$

where the first and last term vanish due to the cross-terms $\tilde{y}_i^\top \tilde{x}_{i'}$ which vanish for any i, i'. Thus, W and W_* encode the same associations, when d and d' are large enough to ensure near-orthogonality.

Layer-normalization. Normalization layers [5, 62] are typically used in transformers to improve training stability [53], and are applied on each token representation, either after [53] or before [7] each block. It may be seen as an operation of the form[4]

$$LN(x) = \frac{x}{\|x\|},$$

applied to the input or output of a given block.

In order to obtain a basic understanding of the role of layer-norm in our associative memory setup, we may consider the setup Lemma 1, with a normalization applied after the linear operation, leading to the population loss:

$$L(W) = \mathbb{E}_{(x,y)\sim p}[\ell(y, W_U LN(Wx))]. \tag{37}$$

The gradients then take the form

$$\nabla_W L(W) = \sum_{k=1}^N \mathbb{E}_x \left[\frac{\hat{p}_W(y=k|x) - p(y=k|x)}{\|Wx\|} \left(I - \frac{(Wx)(Wx)^\top}{\|Wx\|^2} \right) w_U(k) x^\top \right]. \tag{38}$$

[3] When d' is the head dimension, the $\frac{d}{d'}$ scaling can be interpreted as the correct multiplier to use in attention logits, which plays a similar role to the $\frac{1}{d'}$ multiplier in the μP scaling [61], for our setup where the variance of the random entries of input embeddings is $1/d$ instead of 1 as in [61].

[4] RMSNorm [62] would use the variance instead of the norm in the denominator, leading to an additional $\sqrt{d}$ factor in the numerator. Here we use the norm, which is more natural when embeddings have near-unit norm, in contrast to the $\approx \sqrt{d}$ norm for the standard parameterization.